

CS201 Midterm 1 Part A

20 March 2024

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I. QUESTION (ALGORITHM ANALYSIS)

```

1  int algorithm1(int N){
2      if (N == 0)
3          return 0;
4      int sum = 0;
5      for (int i = 0; i < N; i++)
6          for (int j = 0; j < N; j++)
7              for (int k = 0; k < N; k++)
8                  sum++;
9      for (int i = 0; i < 4; i++)
10         for (int j = 0; j < 4; j++)
11             if ((i + j) % 2 == 0)
12                 sum += algorithm1(N / 2);
13             else
14                 sum++;
15     return sum;
16 }
```

What is the time complexity of algorithm1? Solve the recurrence equation without the master theorem.

II. QUESTION (ALGORITHM ANALYSIS)

```

1  int algorithm2(int N){
2      int i, j, k, sum = 0;
3      for (i = 1; i <= N; i++)
4          sum++;
5      for (i = 1; i <= N; i++)
6          for (j = N - i; j > i; j--)
7              for (k = 1; k <= j - i; k++)
8                  sum++;
9      for (i = 0; i <= N; i++)
10         for (j = i + 1; j <= N - i; j++)
11             sum++;
12     return sum;
13 }
```

What does the function algorithm2 return in terms of N ? Assume that N is even.